Alex Rivera

Firmware Engineering Manager | Motor Control, Hardware-in-the-Loop Testing and Launches

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Summary

Firmware engineering manager and principal engineer for warehouse robots. Managed a team of 7 firmware and test engineers, hired 4, and was software lead on 3 robot launches.

Skills

- **Firmware:** C++, FreeRTOS, CAN, motor control, PID loops, hardware-in-the-loop testing
- **Simulation and tooling:** Python, drive and battery simulation, comparison against logged runs
- **AI-assisted development:** Claude Code, coding-agent workflows, automated gates, LLM evaluation
- **Leadership:** team management, hiring, launches

Experience

Principal Firmware Engineer Jan 2023 – Oct 2026
Northbridge Robotics | Madison, WI

- Built an internal coding-agent workflow with Claude Code, reviewed by automated gates, that delivered 180 tasks.
- Measured it in a pilot of 8 engineers over 6 weeks. Median review time fell about 25%.
- Wrote an LLM evaluation harness in Python and ran it against 300 past pull requests.
- Replaced a hand-tuned PID loop with a calibrated model, cutting settling time about 35% on the test rig.
- Built a simulation of drive, battery and wheel slip, compared against 60 logged runs.

Firmware Engineering Manager Mar 2020 – Jan 2023
Northbridge Robotics | Madison, WI

- Scope: a team of 7 firmware and test engineers. Hired 4 engineers, and 2 were promoted.
- Software lead on 3 robot launches.
- Built a hardware-in-the-loop bench with 212 automated tests, run on every merge.

Senior Firmware Engineer Jun 2017 – Mar 2020
Northbridge Robotics | Madison, WI

- Wrote motor-control firmware in C++ for a fleet of 1,400 warehouse robots.
- Cut motion-controller boot time from 42 s to 9 s.

Embedded Software Engineer Aug 2012 – Jun 2017
Lakeshore Instruments | Milwaukee, WI

- Wrote embedded software for industrial instruments.

Education

- B.S., Computer Engineering, University of Wisconsin–Madison, 2012

Certifications

- Certified ScrumMaster